Progress report, 5 October 2026: chord number is chaos degree, and the decoder's fourth moment is free liberation plus localization

`syk_gravity` project

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Abstract

The 29 September report [1] closed with three "next computations" (its §7.3). This report answers the first and most of the second. Throughout, P is the BPS projector of one-flavor $\mathcal{N}{=}2$ SYK ($p = 3$), the decoder is $O = P(1 - n_i)P$, and $P' = UPU$ with $U = (-1)^{n_i}$.

- **Chord number is a Wiener-chaos degree** (item 7.3.1). For OU-correlated couplings $\omega_x = x\omega + \sqrt{1-x^2}\,\omega'$, one has exactly $E\,\tau(P(\omega)P(\omega_x)) = \sum_n \pi_n x^{2n}$, with π_n the weight of the n-th chaos of P. This is BLY's chord-number distribution with exact finite-N weights. Size is bounded by chord number, $k \le (p-1)n$ at half filling (plethysm). The chord-1 sector is exactly $\gamma H^{(2)}$, with a closed form that tends to BLY in double scaling.
- **One generating function for decoder and OU pairs** (item 7.3.2). The decoder, the isotropic OU family and a "site-local" family are evaluations of one Mehler series at different coupling correlations.
- **Isotropic deformations look free.** Along the whole OU family, T_4 as a function of T_2 is within ≤ 0.012 of *free liberation* (a free unitary Brownian rotation of P, which has a closed form). The deviation tracks the known two-model non-freeness.
- **The decoder is local.** At equal T_2 the decoder exceeds the OU curve by 0.017–0.047 in T_4. Gaussian resampling of only the couplings touching site i reproduces two thirds of this, and the same family's $\mathbb{Z}_2$ endpoint reproduces it exactly.
- **What R is.** The decoder's excess over the Wachter value splits as 68–78 % free liberation (analytic), ≤ 5 % isotropic non-freeness and $37 \to 19$ % localization ($N' = 8 \to 14$). The remainder R of the 29 September decomposition is a small difference between a negative liberation piece and a positive localization piece.

Consequence for a gravity computation: the universal object is the isotropic four-arc function, an overlap of two mismatched BPS wormholes. The localization piece is a finite-p, site-level effect that structureless matter chords cannot capture. Tags: [literature], [derived], [derived + verified], [numerical], [interpretation], [conjecture].

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1 Summary of results

1. [derived + verified] **Chord number = chaos degree** (§3). $E\,\tau(P(\omega)P(\omega_x)) = \sum_n \pi_n x^{2n}$, with $\pi_n = E\|P_{[n]}\|^2/d$ and $\pi_0 = a$ exactly. In the chord expansion each Wick contraction across the cut carries x. In double scaling this is the second quantization $x^{\hat{N}}$ of Bożejko–Kümmerer–Speicher, applied to SYK by Pluma–Speicher [literature].

2. [derived + verified] **Selection rule** (§4). The joint (chord, size) distribution J_{nk} vanishes unless $k \le n(p-1)+1$, and unless $k \le n(p-1)$ at half filling. This is the finite-p form of BLY's size $= p\,n$: one unit of size per chord pair is lost to the shared (friend) mode.

3. [derived + verified] **Exact chord-1 sector.** $P_{[1]} = \gamma H^{(2)}$, and $\pi_1/\pi_0 = m_1^2/(m_2 - (1 + M^{-1})m_1^2) = 7/27,\ 24/55,\ 220/351,\ 52/63$ at $N' = 8$–14. The ratio tends to BLY's $2q/(1-q)$ as $p \to \infty$, with corrections $\propto p^{-2}$.

4. [derived] **Krylov form.** $\kappa_n = a\,p_n(0)^2$ exactly, and BLY's $P_n = a_\lambda p_n^{\mathrm{DS}}(0)^2$. [numerical] $\kappa_n \approx \pi_n$ to 3–10 % for $n = 2, 3$.

5. [numerical] **Diagonal dominance.** The share of chord sector 2 at size 4 is 0.47, 0.71, 0.82, ≈ 0.87 at $N' = 8$–14. Top-size parts lie along $(H^n)^{(2n)}$, which gives a lower bound saturated to 1–5 %.

6. [derived] **The probe sees size.** The chord-1 sector is weighted by the character $\hat{\chi}_2(U)$, not by the chord rule's x^2 (0.505 vs 0.424 at $N' = 14$). BLY's 2 % agreement with the decoder variance is a compensation of $+0.034$ (at $n = 0$) and -0.028 (at $n = 1$).

7. [derived] **One generating function** (§5). For a diagonal coupling correlation K, $T_4(K) = \sum_\alpha K^\alpha t_\alpha$. The OU family is $K = x\mathbb{1}$, the decoder $K_J = (-1)^{[i\in J]}$, and the site-local family $K_J = z^{[i\in J]}$. The four-arc rule (adjacent-arc chords K_J, opposite-arc chords 1) is the 29 September $\mathbb{Z}_2$ rule. It is *not* the random-matter rule of BINT (opposite chords $\tilde{q}^2$). The decoder equals OU at x_s exactly when cross-chord index sets are uncorrelated, which holds in double scaling for heavy probes.

8. [derived] **Free-liberation baseline** (§6). The free Jacobi process of P against itself has $T_2 = a + (1-a)\eta$ and the closed form $R_{\mathrm{lib}}(\eta) = 2a(1-a)(1-2a)\eta + [E + 2a(1-a)^2 \ln \eta]\eta^2$. It explains the sign changes of R at $a = \frac{1}{2}$ and $a = 0.618$ found on 29 September.

9. [numerical] **OU $\approx$ liberation; decoder $\neq$ OU** (§7). Table 4 gives the three-way split. The localization term is reproduced by the site-local family and is already visible at infinitesimal angle, as excess heterogeneity of the quantum metric (κ 22–49 % above free, against 3–12 % for OU).

10. [numerical] **Matched OU parameter.** $x_* \approx \sqrt{\hat{\chi}_2(U)}$ to within 2 % (3.7 % at $N' = 9$), as item 6 predicts. The naive per-chord value $1 - 2p/N'$ is far off.

11. [derived] **Small-angle limit.** Isotropic OU noise sees only the quantum metric; the Berry curvature cancels. So the decoder's T_4 is a quantum-metric quantity, not the curvature moment of Chen et al. or G. Lin.

2 Setup, recap, and what is new

Model. One-flavor $\mathcal{N}$=2 SYK on $N \equiv N'$ complex fermions, with $Q = \sum_{|I|=p} C_I \psi_I$, $p = 3$, and $H = \{Q, Q^\dagger\}$.

- The couplings $\omega = (C_I) \in W = \Lambda^p \mathbb{C}^N$, $M = \binom{N}{p}$, are i.i.d. standard complex Gaussians.
- We work in the charge sector Λ^F, $F = \lfloor N/2 \rfloor$, $D = \binom{N}{F}$. P is the BPS projector, $d = \mathrm{Tr}\, P$, $a = d/D$, $\tau = \mathrm{Tr}\,/d$.
- The BPS dimension is BBNR's finite-N cohomology count [4] (their eq. 5.25), $d = \sum_n (-1)^n \binom{N}{F+np}$. At $p = 3$ this is $2 \cdot 3^{N/2-1}$ at half filling and $3^{(N-1)/2}$ at odd N [literature].

Decoder. $O = P(1-n_i)P = \frac{1}{2}(P + PUP)$ with $U = (-1)^{n_i}$, and $(PUP)^2 = PP'P$ with $P' = UPU$. The moments are $T_{2m} = \tau((PP')^m)$, so $T_2 = \tau(PP')$ and $T_4 = \tau(PP'PP')$.

Recap of 29 September [1].

- Exact sum rule: $E\,T_2 = \sum_k w_k \hat{\chi}_k(U)$, with w_k the $U(N)$ size spectrum of P, $w_0 = a$, and $\hat{\chi}_k(U) = 1 - 4k(N+1-k)/(N(N+1))$.
- Two independent models: $E\,\tau(P_1P_2P_1P_2) = \sum_k \phi_k w_k \equiv c_0$. SYK is +0.6, +2.0, +4.3 % above free at $N' = 10, 12, 14$.
- Decoder: $T_4 = a^2 + 2a(T_2 - a) + S + R$ exactly, with $S = c_0 - a^2$. The remainder R was open (about 21 % of the excess at $N' = 13$–14).
- The $\mathbb{Z}_2$ arc rule: a Q-chord separated from its partner by an odd number of U's carries $(-1)^{[i\in I]}$.

What is new.

- §§3–4: the derivation asked for in item 7.3.1 (full note: `research/notes/chord_size_dictionary_2026-10-02.md`; summary D4 in `docs/derivations.md`).
- §§5–8: item 7.3.2, preceded by a literature step. Five papers were added to `papers/` with notes; see Appendix B. (Full note: `research/notes/decoder_T4_fourarc_2026-10-05.md`; derivation D5.)

3 Chord number is a Wiener-chaos degree

3.1 Definition and the Mehler formula [derived + verified]

$P(\omega)$ is a square-integrable $\mathrm{End}(\Lambda^F)$-valued function of the Gaussian vector ω, so it has a Wiener-chaos (Wick) expansion. Phase invariance, $P(e^{i\theta}\omega) = P(\omega)$, leaves only bidegrees (n, n):

$$P = \sum_{n\geq 0} P_{[n]}, \qquad \pi_n = \frac{E\|P_{[n]}\|_F^2}{d}, \qquad \sum_n \pi_n = 1. \tag{1}$$

The Ornstein–Uhlenbeck semigroup $T_x f(\omega) = E_{\omega'} f(x\omega + \sqrt{1-x^2}\omega')$ acts on bidegree (n, n) as x^{2n}. Hence

$$G(x) \equiv E\,\tau\big(P(\omega)P(\omega_x)\big) = \sum_n \pi_n\, x^{2n}, \qquad \frac{E\langle P^{(k)}(\omega), P(\omega_x)\rangle}{d} = \sum_n J_{nk}\, x^{2n}, \tag{2}$$

where $J_{nk} = E\|P^{(k)}_{[n]}\|^2/d$ is the joint (chord, size) distribution. It exists because the chaos and size gradings both commute with $U(N)$, and its marginals are π_n and w_k.

- $n = 0$: $P_{[0]} = E[P] = a\mathbb{1}$ by Schur's lemma, so $\pi_0 = w_0 = J_{00} = a$ *exactly.* This is the exact form of BLY's "probability of zero length = BPS fraction" [3], with the exact a in place of the double-scaled a_λ.
- **Chord reading.** Expand both projectors as $\lim e^{-\beta H}$ and Wick-contract. Each contraction joining the two copies carries x; internal contractions carry 1. The coefficient of x^{2n} collects the diagrams with n Q- and n $Q^\dagger$-chords crossing the cut, which are BLY's n XO pairs. So (2) is BLY's (4.5)–(4.6) with $q^\Delta \to x$, with exact finite-N Wick weights in place of double-scaled chord rules.

3.2 Double scaling: second quantization [literature] + [derived]

In the double-scaled Majorana model, the two statements above are rigorous theorems in the literature.

- Independent SYK copies converge to q-Gaussians $s_q(h_i)$ with orthonormal h_i (Pluma–Speicher [6], Thm. 3.1). By the multilinearity argument of their Cor. 3.2, correlated couplings give $\langle h, h_x\rangle = x$.
- Conditioning on one copy is the second quantization Γ_q of a projection (Bożejko–Kümmerer–Speicher [7], Thm. 2.11). It acts as $x^{\hat{N}}$ on the chord Fock space.
- Hence $E\ \mathrm{tr}\, f(H)g(H_x) = \langle f|x^{\hat{N}}|g\rangle$. Its kernel between energy eigenstates is Rogers' q-Mehler kernel (BKS Thm. 1.10), which is exactly BINT's matter vertex (5.4) with $\tilde{q} \to x$ [5].

So "an OU correlation x" and "a matter chord of weight $\tilde{q} = q^{\Delta}$" are the same operation on chord space [derived]. This also corrects an earlier project note, which called correlated copies "beyond Bożejko–Speicher".

3.3 Krylov form [derived]

With $\langle X, Y\rangle = E\ \mathrm{tr}(X^\dagger Y)$, $f \mapsto f(H)$ is an isometry from $L^2(\mu)$, where μ is the averaged density of states. μ is determinate and has an atom a at $E = 0$. Hence the Krylov weights of $P = \mathbb{1}_{\{0\}}(H)$ are

$$\kappa_n = a\, p_n(0)^2, \qquad \sum_n \kappa_n = 1, \tag{3}$$

with p_n the orthonormal polynomials of μ. In BLY's C-even chord basis H is tridiagonal, so $P_n = a_\lambda\, p_n^{\mathrm{DS}}(0)^2$ exactly. At finite N the chaos and Krylov gradings differ:

- $\kappa_0 = \pi_0$, and $\kappa_1 = \pi_1$ up to $O(1/M)$;
- [numerical] κ_2, κ_3 are within 3 % and 10 % of π_2, π_3 (Table 2).

4 Size is bounded by chord number

4.1 Selection rule [derived + verified]

Proposition 1. *For odd p, $P^{(k)}_{[n]} = 0$ unless $k \le n(p-1)+1$. At half filling, $P^{(k)}_{[n]} = 0$ unless $k \le n(p-1)$. For $p = 3$ the joint distribution is supported on $k \le 2n$.*

Proof sketch.

1. $P^{(k)}_{[n]}$ is an invariant vector in $\mathcal{H}_{n,n} \otimes V_k$, with $\mathcal{H}_{n,n} \simeq \mathrm{Sym}^n W^* \otimes \overline{\mathrm{Sym}^n W^*}$.
2. In the stable range, $V_k = [1^k; 1^k]$ occurs in $V_\mu \otimes \overline{V_\nu}$ only if μ contains a vertical k-strip (Koike [15]), so $k \le \ell(\mu)$.
3. For odd p, $h_n[e_p] = \omega(e_n[h_p])$ (Macdonald [16]). Its longest row has length $p + (n-1)(p-1)$, attained by $x_1^p \wedge x_1^{p-1} x_2 \wedge \cdots$. So $\max \ell(\mu) = n(p-1)+1$. Checked by explicit Schur expansion for $(n,p) = (2,3), (3,3), (2,5)$.
4. At half filling particle–hole parity kills odd k (residual $\le 10^{-16}$ per realization). Since $n(p-1)+1$ is odd, $k \le n(p-1)$. □

Reading [interpretation].

- The couplings commute, but for odd p the operators ψ_I anticommute. Configurations in which the n chords touch pairwise-disjoint modes cancel, and at least $n-1$ coincidences are forced (consecutive chords are friends).
- For $n=1$ this is explicit. $\{\psi_I, \psi_J^\dagger\} = 0$ for disjoint I, J, and for $I \cap J = \{s\}$ it equals $\pm\psi_{I\setminus s}\psi_{J\setminus s}^\dagger$. So H has no size-p part at any filling.
- BLY's (5.23), $\text{size}_O = p\, n_O$, is the leading-order statement in $1/p$.

Verified [numerical].

- $J_{1,4} = J_{1,6} = 0$ for all N', and $J_{2,6} = 0.0000(1)$ at $N' = 12$, while $J_{3,6} = 0.0077$.
- Off half filling the general bound is attained: $J_{1,3} = 0.0035$ at $N' = 9$, $F = 4$, which is 279σ from zero.

4.2 The exact chord-1 sector [derived + verified]

The chaos-1 space $\Lambda^3 \otimes \overline{\Lambda^3} = \bigoplus_{r=0}^3 V_r$ is multiplicity-free, and parity and $P^{(0)} = a\mathbb{1}$ exclude $r \neq 2$. $H^{(2)}$ is a pure chaos-1 invariant. Hence $P_{[1]} = \gamma H^{(2)}$. From $PH = 0$ and $\operatorname{Tr} H \propto |\omega|^2$,

$$\frac{\pi_1}{\pi_0} = \frac{m_1^2}{m_2 - (1 + M^{-1})\, m_1^2}, \qquad m_k = E\ \operatorname{tr} H^k. \tag{4}$$

In double scaling ($|I \cap J|$ Poisson($\lambda/2$)) this tends to $2q/(1-q)$, which is BLY's P_1/P_0 at $j_R = 0$ (their 3.6–3.10). An independent check is the chaos-$(1,0)$ projection of $QP = 0$, which gives the same γ to all printed digits.

N'	π_1/π_0, eq. (4)	BLY $2q/(1-q)$	π_1 (exact)	π_1 (OU fit, §7)	BLY P_1	w_2
8	$7/27 = 0.2593$	0.2356	0.2000	0.2005(6)	0.1860	0.2147
10	$24/55 = 0.4364$	0.3961	0.2805	0.2802(7)	0.2657	0.3004
12	$220/351 = 0.6268$	0.5744	0.3297	0.3298(8)	0.3209	0.3510
14	$52/63 = 0.8254$	0.7642	0.3506	0.3503(7)	0.3506	0.3706

Table 1: [derived + verified] The exact chord-1 weight. The OU-fit column comes from today's independent campaign (§7). At fixed λ, the relative deviation from BLY is $\approx c(\lambda)/p^2$: 6.3 % at $p = 3$, 1.1 % at $p = 7$, 0.04 % at $p = 35$. So chord weights at $p = 3$ carry 6–10 % finite-p corrections. $w_2 - \pi_1 = \sum_{n\geq 2} J_{n2} \geq 0$ is the leakage of higher chord sectors into size 2.

4.3 The joint distribution and the top-size bound

Top-size bound [derived]. $X_n \equiv (H^n)^{(2n)}$ is a pure chaos-n, size-$2n$ invariant vector, so by Cauchy–Schwarz

$$J_{n,2n} \;\geq\; \frac{\left(E\langle P, X_n\rangle\right)^2}{d\, E\|X_n\|^2}. \tag{5}$$

N'	$J_{2,2}$	$J_{2,4}$	bound (5)	$J_{2,4}/\pi_2$	$J_{3,6}$	π_2 / κ_2 / P_2^{BLY}	π_3 / κ_3 / P_3^{BLY}
8	0.0132	0.0115	0.0115	0.47	–	0.0246 / 0.0249 / 0.0219	0.0033 / 0.0038 / 0.0023
10	0.0177	0.0431	0.0426	0.71	–	0.0608 / 0.0592 / 0.0526	0.0122 / 0.0137 / 0.0090
12	0.0191	0.0858	0.0851	0.82	0.0077	0.1050 / 0.1013 / 0.0922	0.0280 / 0.0302 / 0.0218
14	≈ 0.018	0.128–0.132	0.1284	≈ 0.87	≥ 0.0230	– / 0.1437 / 0.1339	– / 0.0521 / 0.0407

Table 2: [numerical] Joint distribution from OU campaigns with exact Casimir sizes. There were 4000, 744 and 174 samples at $N' = 8, 10, 12$ ($K = 16$ angles each). $N' = 14$ uses top-size projections (40 realizations) plus a 5-sample OU run. Coefficients with $n \leq 3$ are stable to $\lesssim 10^{-4}$ under split halves and cutoffs; $n \geq 4$ are not quoted. The bound (5) is saturated to 1 % ($n = 2$) and about 5 % ($n = 3$). The exact chord distributions are fatter-tailed than BLY's at $p = 3$.

4.4 Consequences for the 29 September dictionary [derived]

- **The empirical match $w_{2n} \approx P_n$ (10–20 %) decomposes exactly** as $w_{2n} - P_n = (w_{2n} - \pi_n) + (\pi_n - P_n)$. The first term is size leakage between chord sectors, which vanishes for $n = 0$. The second is finite-p chord weights, dominated by a vs a_λ; the conditional shapes agree to a few per cent.
- **The probe sees size, not chord number.** Under U the couplings rotate by $\Lambda^p U$. On chaos n this acts as its second quantization, whose average over the size-k part is $\hat{\chi}_k(U)$, so $T_2 = \sum_{nk} J_{nk} \hat{\chi}_k(U)$.
  - The chord-1 sector is weighted by $\hat{\chi}_2(U) = 1 - 8(N-1)/(N(N+1))$: 0.222, 0.346, 0.436, 0.505 at $N' = 8$–14. The double-scaled $e^{-12/N'}$ gives 0.223, 0.301, 0.368, 0.424, and the naive $(1 - 6/N')^2$ gives 0.063–0.327.
  - Reason: friend chords share a mode, on which the probe's sign cancels. The probe therefore sees $2(p-1)$ modes per chord pair rather than $2p$.
  - This resolves item 3 of the 29 September open list: per-chord weights do not factorize, and the exact pair weight is a character.
- **BLY's 2 % variance agreement is a compensation.** At $N' = 14$, $T_2 = a + \pi_1 \hat{\chi}_2 + \sum_{n \geq 2}$ $= 0.4248 + 0.1770 + 0.0353$. BLY's terms are 0.4588, 0.1488 and 0.0276: $+0.034$, -0.028, -0.008.

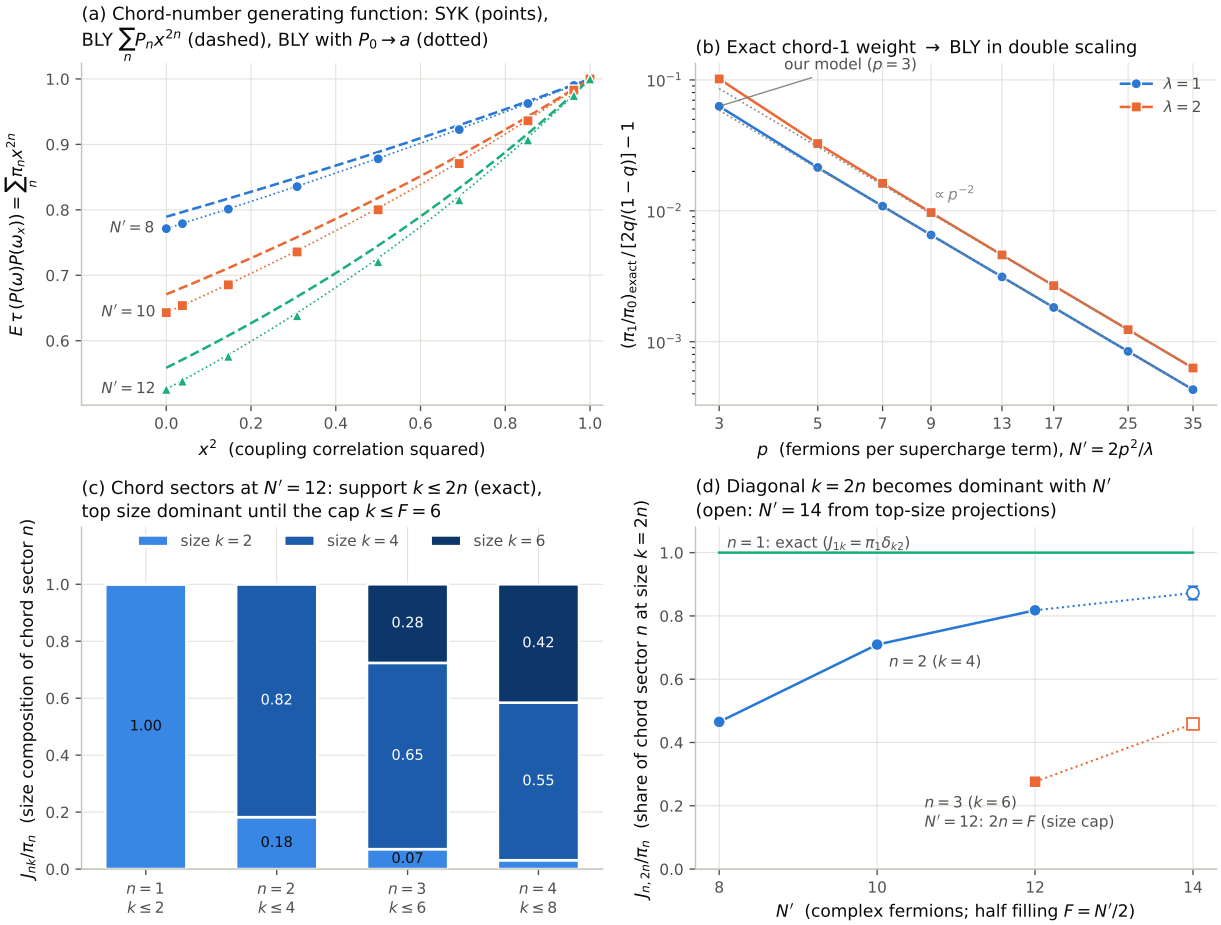


Figure 1: The chord-number/size dictionary.

- (a) [derived + verified] Generating function (2) from angle-rotated OU pairs (points; errors smaller than markers). Dashed: BLY's $\sum_n P_n x^{2n}$ at $\lambda = 18/N'$. Dotted: the same with $P_0 \to a$.
- (b) [derived + verified] The exact chord-1 weight (4) relative to BLY at fixed λ, approaching it as p^{-2}.
- (c) [numerical] Size composition J_{nk}/π_n at $N' = 12$. The support $k \le 2n$ is exact; $n = 4$ is shown for completeness only (cutoff-sensitive).
- (d) [numerical] The share of the top size $k = 2n$. Open symbols at $N' = 14$: top-size projections (5).

5 The fourth moment as a four-arc function

5.1 One generating function for three coupling families [derived]

Let $K = \mathrm{diag}(K_J)$ be a real diagonal correlation on coupling space, $|K_J| \le 1$, and let $\tilde{C} = KC + \sqrt{1-K^2}\,C'$ with C' an independent copy. Define

$$T_{2m}(K) = E\,\tau\big((P(C)P(\tilde{C}))^m\big). \tag{6}$$

Three choices of K concern us (Fig. 2c):

- **OU family**, $K_J \equiv x$: all couplings partially resampled.
- **Decoder**, $K_J = \sigma_J \equiv (-1)^{[i\in J]}$: deterministic. This is exact because $UQ(C)U = Q(\sigma C)$, hence $UP(C)U = P(\sigma C)$ (tested to 10^{-10}). The parity family U_A has $\sigma_J = (-1)^{|A\cap J|}$.
- **Site-local family**, $K_J = z$ for $J \ni i$ and 1 otherwise. Only the couplings touching site i are resampled. Then $z = 0$ is a fresh draw of those couplings, and $z = -1$ is the decoder.

Writing $F(C) = P(C)\otimes P(C)$ and $\mathcal{B}(A\otimes B, C\otimes D) = \tau(ACBD)$, the multivariate Mehler formula (valid also at $|K_J| = 1$, by parity of Hermite polynomials) gives

$$\boxed{T_4(K) = \sum_\alpha K^\alpha\, t_\alpha, \qquad t_\alpha = \mathcal{B}(F_\alpha, F_\alpha)} \tag{7}$$

Here α runs over multi-indices of Wick contractions between arcs of different flavour. **The three families evaluate the same coefficients t_α at different points K.** Unlike π_n, the t_α need not be positive: $T_4 = \tau(Z^2)$ with $Z = P(C)P(\tilde{C})$, whereas the time-ordered $\tau(Z^\dagger Z)$ is a norm.

5.2 The four-arc rule, and why it is not random matter [derived]

Contractions between the two C-arcs are internal to F_α and carry 1. So on a circle with arcs P, P', P, P':

- chords between *adjacent* arcs carry K_J;
- chords between *opposite* arcs carry 1.

This is the 29 September $\mathbb{Z}_2$ rule, now with an exact finite-N realization by Gaussian couplings (Fig. 2a). For a fixed $\mathbb{Z}_2$ probe all three pairings of the four U's give the same sign, $(-1)^{\#U \text{ between the chord ends}}$, so there is no sum over U-pairings.

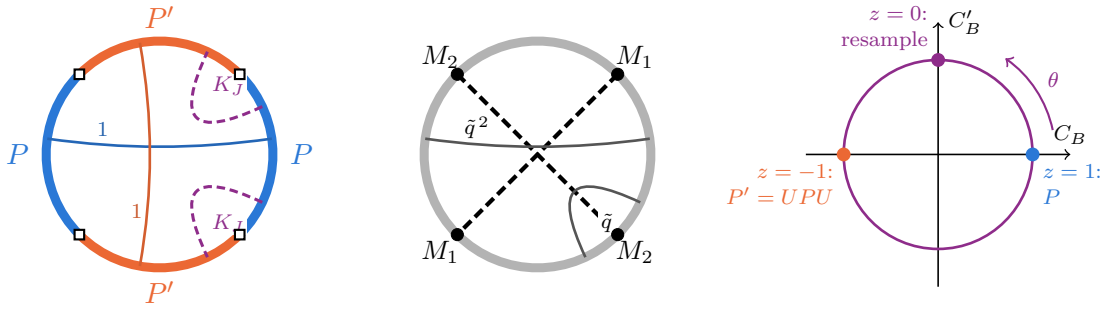
For BINT's random matter M (independent couplings, crossing weight $\tilde{q} = e^{-2pp_M/N}$ per H-chord), an H-chord between opposite arcs crosses both M-chords and carries $\tilde{q}^2$ (Fig. 2b; BINT eqs. 2.9–2.11, 4.10–4.13 [5]). The crossed four-point function and its $U_q(\mathfrak{su}(1,1))$ R-matrix (their 5.6) therefore do *not* give T_4 by substituting $\tilde{q} = x$. What is needed is the crossing kernel of the $\mathbb{Z}_2$ rule, which is not in the literature we have.

5.3 When is the decoder an OU pair? [derived]

By S_N invariance the site-averaged decoder weight of a multi-index α is

$$E_i[\sigma^\alpha] = 1 - \frac{2\,o(\alpha)}{N}, \qquad \text{more generally} \quad E_{|A|=s}[\sigma^\alpha] = \frac{\mathcal{K}_s(o(\alpha))}{\binom{N}{s}}, \tag{8}$$

where $o(\alpha)$ is the number of sites covered an odd number of times by the cross-chords' index sets (the support of the monomial $\prod\psi$) and $\mathcal{K}_s$ is a Krawtchouk polynomial.



(a) $\mathbb{Z}_2$ / OU rule: adjacent K_J, opposite 1

(b) random matter (BINT): adjacent $\tilde{q}$, opposite $\tilde{q}^2$

(c) site-local family: only C_J, $J \ni i$, rotate; $z = \cos\theta$

Figure 2: [derived] Four-arc chord rules and coupling families.

- (a) In $T_4 = \tau(PP'PP')$, chords between adjacent arcs carry the coupling correlation K_J: x for OU pairs, $(-1)^{[i\in J]}$ for the decoder. Chords between opposite arcs carry 1.
- (b) For BINT's random matter, an H-chord between opposite arcs crosses both matter chords and carries $\tilde{q}^2$. The two rules differ, so the random-matter R-matrix does not apply.
- (c) The site-local family rotates only the couplings touching site i in the plane (C_B, C'_B). Its antipode is exactly the decoder. The OU family rotates all couplings in the plane (C, C'), with $x = \cos\theta$ and $P(-C) = P(C)$.

- For *independent uniformly random* index sets, $E[\sigma^\alpha] = x_s^{|\alpha|}$ exactly, with $x_s = \mathcal{K}_s(p)/\binom{N}{s}$. So *the decoder equals OU at x_s, coefficient by coefficient, exactly when the cross-chord index sets are uncorrelated.*
- **Double scaling with a heavy probe** ($s \propto \sqrt{N}$, p^2/N fixed). Overlap correlations shift o by $O(1)$ and change the weights by $O(s/N) \to 0$, so decoder $\equiv \mathrm{OU}(x_s)$ for every T_{2m} at leading order. A single-site probe trivializes there ($x_s \to 1$).
- **Finite p.** The correlations are large: the chord-1 weight is $\hat{\chi}_2(U)$, not x_s^2 (§4). The meaningful comparison is therefore at the *matched* x_* defined by $T_2^{\mathrm{OU}}(x_*) = T_2^{\mathrm{dec}}$. At chaos order ≤ 1 this predicts $x_* \approx \sqrt{\hat{\chi}_2(U)}$.

5.4 Exact properties of the OU curve [derived]

- $T_2^{\mathrm{OU}} = \sum_n \pi_n x^{2n}$ and $T_4^{\mathrm{OU}} = \sum_n \theta_n x^{2n}$, even since $P(-C) = P(C)$. $T_4(1) = 1$ and $T_4(0) = c_0$.
- **Sum rule.** $\sum_n n\,\theta_n = 2\sum_n n\,\pi_n$, since $\partial_x T_4 = 2\partial_x T_2$ at $x = 1$ by $P^2 = P$.
- **Remainder.** With $Y_x = P_x - a$, the split $Q_4 = S + R$ of 29 September applies along the curve. At $x = 0$, $E_{C'}|Y'\rangle\rangle\langle\langle Y'|$ is $U(N)$-invariant and hence orthogonal to $\delta\mathcal{P}$, so $R^{\mathrm{OU}}(0) = 0$ exactly. Also $R^{\mathrm{OU}}(1) = (1-a)^2 - S$.
  - [numerical] $S^{\mathrm{OU}}(x) = E\sum_k \phi_k(\omega) w_k(\omega_x)$ depends on x only through covariances of per-realization size data. These are $\leq 8\times 10^{-7}$ at $N' = 8$–14, so S is effectively x-independent.
- $\sin^4$ **identity.** $T_4 - 2T_2 + 1 = \tau\big((PP_xP - P)^2\big) = E\sin^4\theta$ over principal angles. For the decoder this is $4\tau((PP_oP)^2)$ with $P_o = (P - UPU)/2$.

- **Small angle** ($\epsilon = 1 - x$, real coordinates ω_I of unit variance). $PP_xP - P = -B^\dagger B + O(\delta\omega^3)$ with $B = \sum_I \delta\omega_I (1-P)\partial_I P\, P$. Hence

$$1 - T_2 \simeq \frac{2\epsilon}{d} E\ \mathrm{Tr}\, g_{\mathrm{tot}}, \qquad T_4 - 2T_2 + 1 \simeq \frac{4\epsilon^2}{d} E\Big[\ \mathrm{Tr}\, g_{\mathrm{tot}}^2 + 2\sum_{IJ} \mathrm{Tr}\, g_{IJ}^2\Big]. \tag{9}$$

  Here $G_{IJ} = P\partial_I P(1-P)\partial_J P\, P$ is the operator-valued quantum geometric tensor (Chen et al. [12], App. A), $g_{IJ} = (G_{IJ} + G_{JI})/2$ and $g_{\mathrm{tot}} = \sum_I g_{II}$.

  - The Berry curvature $F_{IJ} = G_{IJ} - G_{JI}$ cancels between the $IJIJ$ and $IJJI$ Wick orderings.
  - So isotropic noise measures the variance of the quantum metric, not the curvature moment $\mathrm{Tr}\, FF^\dagger$, which G. Lin writes as TOC−OTOC [11] (his 6.15–6.19).
  - The ratio $\kappa \equiv (T_4 - 2T_2 + 1)/(1 - T_2)^2 \to d\, E[\cdots]/(E\ \mathrm{Tr}\, g_{\mathrm{tot}})^2$ is the normalized heterogeneity of the BPS states' sensitivity.

- **Conjugation flow.** $P(\theta) = e^{i\theta n_i} P e^{-i\theta n_i}$ interpolates $P \to P'$, a conjugation deformation in the sense of [12]. Its quantum metric is $\hat\Lambda(1-\hat\Lambda)$ with $\hat\Lambda = Pn_iP = P - O$, and $T_2(\theta) = 1 - 2(1-\cos\theta)\,\tau(\hat\Lambda - \hat\Lambda^2)$ exactly.

5.5 Two-sided reading [derived] (double-scaled Majorana) + [interpretation] ($\mathcal{N} = 2$)

In the q-Gaussian GNS space left and right multiplications commute. Hence

$$T_4^{\mathrm{OU}} = \langle \Psi_{h_x,h} | \Psi_{h,h_x} \rangle, \qquad |\Psi_{h,h'}\rangle = L\big(P(h)\big)\, R\big(P(h')\big)\, \Omega, \tag{10}$$

- $|\Psi_{h,h'}\rangle$ is the zero-energy state of the two-sided chord Hilbert space [10]. Its left boundary evolves with $H(\omega)$ and its right boundary with $H(\omega_x)$. Chords are coloured by their emitting boundary, and the inner product is the q-Fock product with $\langle h, h_x\rangle = x$.
- So T_4^{OU} **is the overlap of a mismatched BPS wormhole with its mirror**, and $T_2^{\mathrm{OU}} = \langle \Omega | \Psi_{h,h_x}\rangle$.
- By BKS (§4), $(P(\omega), P(\omega_x))$ are values of the q-OU process at two times. T_2 is time-ordered, but T_4 has operator times (s,t,s,t): it is an out-of-time-order correlator of the q-OU process, not fixed by its Markov kernel.
- At finite N this identity is just vectorization. Its content is that the double-scaled limit is a two-colour chord computation, not a matter-chord R-matrix.
- The $\mathcal{N} = 2$ version (BBNR's oriented chords) follows by the same multilinearity, but we have not seen it written down.

6 The free-liberation baseline

The natural Hilbert-space-isotropic null model rotates P against itself by a free unitary Brownian motion Y_t, free from P: the free Jacobi process $J_t = PY_tPY_t^*P$ with $\tau(P) = a$, i.e. Demni–Hamdi–Hmidi parameters $\lambda = 1$, $\theta = a$ [8]. Their moment equation (eq. 2) is stated for any initial data; here $m_n(0) = 1$. With $u = e^{-t}$ and $b = 1 - a$ it gives [derived]

$$T_2 = a + b\,u \quad \Rightarrow \quad \eta \equiv \frac{T_2 - a}{1 - a} = e^{-t} \text{ exactly},$$
$$T_4 = a^2(2-a) + 4ab^2 u + (E - 2ab^2 t)\, u^2, \qquad E = 1 - a^2(2-a) - 4ab^2. \tag{11}$$

Relative to the free two-model value $c_0^{\text{free}} = a^2(2-a)$ (i.e. $S_{\text{free}} = a^2(1-a)$):

$$\boxed{R_{\text{lib}}(\eta) = 2a(1-a)(1-2a)\,\eta + \big[E + 2a(1-a)^2 \ln\eta\big]\eta^2} \tag{12}$$

- **Checks.**
  - It agrees with the numerically integrated ODE to 10^{-11}, and $t \to \infty$ gives the Wachter value.
  - $R_{\text{lib}}(1) = 1 - 2a + a^3 = (1-a)^2 - S_{\text{free}}$.
  - A finite-D simulation ($D = 800$–2000, unitary Brownian motion from GUE exponentials) reproduces the ODE moments to $\lesssim 2 \times 10^{-3}$.
- **Sign structure.** The linear term changes sign at $a = \frac{1}{2}$. $R(1)$ changes sign at the golden-ratio root $a = (\sqrt{5}-1)/2 = 0.618$ of $a^3 - 2a + 1$. This is the threshold observed empirically on 29 September, where $N' = 8, 10$ ($a > 0.618$) had $R < 0$.
- **Small angle.** $\kappa \to 1/(1-a)$: free-Poisson statistics of $B^\dagger B$ for $B = (1-P)\,dW\,P$.
- **Non-analyticity.** The $\eta^2 \ln\eta$ term is non-analytic. The OU curve is a power series in x^2, so OU can approximate liberation but never equal it.

7 Numerics: decoder vs OU vs free liberation

Design.

- Per sample, $K = 16$ BPS projectors are built on a circle in coupling space (angle trick). For the OU family, $\omega_j = \cos\theta_j\,\omega + \sin\theta_j\,\omega'$ with $\theta_j = j\pi/K$, giving 9 values of x including $x = 0$ exactly. For the site family, only C_B rotates, with $\theta_j = 2\pi j/K$, giving $z \in [-1, 1]$; $z = -1$ *is* the decoder.
- $\cos^2$ of the principal angles come from orthonormal BPS bases; moments T_2–T_{16} and spectra are recorded.
- On the same realizations the decoder (all N' single sites) and parity probes ($|A| = 2..N'/2$) are evaluated, so decoder and OU are *realization-matched*.
- Samples: 200, 150, 100, 60, 40, 16, 14 at $N' = 8$–14 (OU), and 150, 80, 30 at $N' = 8, 10, 12$ (site family). Errors are bootstrap over samples.
- **Cross-checks.**
  - The fitted π_1 reproduces (4) (Table 1).
  - The measured endpoints $c_0 = T_4^{\text{OU}}(0)$, namely 0.56428(50), 0.41581(71), 0.29420(126), 0.29744(52) at $N' = 10, 12, 13, 14$, reproduce the 29 September parameter-free two-model predictions 0.56399, 0.41606, 0.29481, 0.29661.
  - R_{dec} agrees with the independent 29 September run: +0.0121 vs +0.0119 at $N' = 12$, +0.0515 vs +0.0500 at 13, and +0.0508 vs +0.0531 at 14.

7.1 Isotropic deformations are close to free liberation [numerical]

At infinitesimal angle the heterogeneity ratio κ of OU is 4.51, 3.02, 2.36 at $N' = 8, 10, 12$. The free values $1/(1-a)$ are 4.38, 2.80, 2.11, so OU is +3, +8, +12 % above free.

N'	a	$q = e^{-18/N'}$	$\max_x(T_4^{\mathrm{OU}} - T_4^{\mathrm{lib}})$ [at x]	same for T_8	$c_0 - c_0^{\mathrm{free}}$
8	0.7714	0.105	−0.0025 [0.383]	−0.0028	−0.0010
9	0.6429	0.135	+0.0043 [0.831]	+0.0099	+0.0025
10	0.6429	0.165	+0.0031 [0]	+0.0057	+0.0034
11	0.5260	0.195	+0.0081 [0.831]	+0.0182	+0.0079
12	0.5260	0.223	+0.0083 [0]	+0.0165	+0.0080
13	0.4248	0.250	+0.0120 [0.831]	+0.0258	+0.0099
14	0.4248	0.276	+0.0121 [0]	+0.0222	+0.0132

Table 3: [numerical] Largest deviation of the OU curve from free liberation at equal T_2, over the 8 lags. It tracks the two-model non-freeness $c_0 - c_0^{\mathrm{free}}$, and both grow with q, as Liu–Shen's inter-model crossing weight $q_{12} = q$ suggests [9]. The bulk of the spectrum coincides with liberation (Fig. 3d). OU has a soft tail below the free lower edge.

7.2 The decoder exceeds the isotropic curve [numerical]

N'	T_2^{dec}	x_*	T_4^{dec}	$T_4^{\mathrm{OU}}(x_*)$	T_4^{lib}	T_4^{free}	liberation	isotropic	localization
8	0.8174(2)	0.476	0.7778(2)	0.7603(2)	0.7627	0.7311	0.0316 (68 %)	−0.0024 (−5 %)	0.0175(1) (37 %)
9	0.7319(3)	0.557	0.6666(3)	0.6373(3)	0.6354	0.5609	0.0745 (70 %)	+0.0019 (2 %)	0.0293(1) (28 %)
10	0.7456(3)	0.582	0.6790(3)	0.6469(3)	0.6475	0.5609	0.0867 (73 %)	−0.0006 (−1 %)	0.0321(1) (27 %)
11	0.6690(3)	0.630	0.5799(4)	0.5419(3)	0.5356	0.4078	0.1278 (74 %)	+0.0064 (4 %)	0.0380(2) (22 %)
12	0.6861(3)	0.650	0.5964(3)	0.5548(3)	0.5519	0.4078	0.1441 (76 %)	+0.0029 (2 %)	0.0417(2) (22 %)
13	0.6184(5)	0.684	0.5101(5)	0.4669(5)	0.4562	0.2843	0.1719 (76 %)	+0.0107 (5 %)	0.0432(2) (19 %)
14	0.6368(4)	0.696	0.5283(4)	0.4816(4)	0.4742	0.2843	0.1900 (78 %)	+0.0074 (3 %)	0.0467(2) (19 %)

Table 4: [numerical] The decoder ($s = 1$) at equal T_2. $T_4^{\mathrm{free}} = 2a^2 - a^3$ (Wachter), and T_4^{lib} is free liberation at T_2^{dec}. The excess $T_4^{\mathrm{dec}} - T_4^{\mathrm{free}}$ splits into liberation $= T_4^{\mathrm{lib}} - T_4^{\mathrm{free}}$, isotropic $= T_4^{\mathrm{OU}}(x_*) - T_4^{\mathrm{lib}}$ and localization $= T_4^{\mathrm{dec}} - T_4^{\mathrm{OU}}(x_*)$. Parentheses: bootstrap SE in units of 10^{-4}.

- The localization term grows but saturates in absolute size (0.017, 0.032, 0.042, 0.047 at even $N' = 8$–14), while its share falls from 37 to 19 %.
- T_6 and T_8 show the same pattern with larger excesses. At $N' = 14$: T_6 0.469 vs 0.394, T_8 0.430 vs 0.337. The decoder's principal angles are more heterogeneous.
- **Matched OU parameter.** $x_* = 0.476, 0.557, 0.582, 0.630, 0.650, 0.684, 0.696$ against $\sqrt{\hat{\chi}_2(U)} = 0.471, 0.537, 0.588, 0.628, 0.660, 0.688, 0.711$ ($N' = 8$–14). The naive $x_s = 1 - 2p/N'$ is 0.25–0.57.
- **Parity family.** For $|A| = s > 1$ the excess is smaller in absolute terms but larger *relative* to the OU distance from independence: 0.30, 0.43, 0.65 for $s = 1, 2, 3$ at $N' = 12$. Heavier parity probes are thus not "more isotropic" at these sizes. This does not contradict the double-scaling statement of §5, which needs $s/N \to 0$ with $p \propto \sqrt{N}$.

7.3 The excess is localization, not the $\mathbb{Z}_2$ sign [numerical]

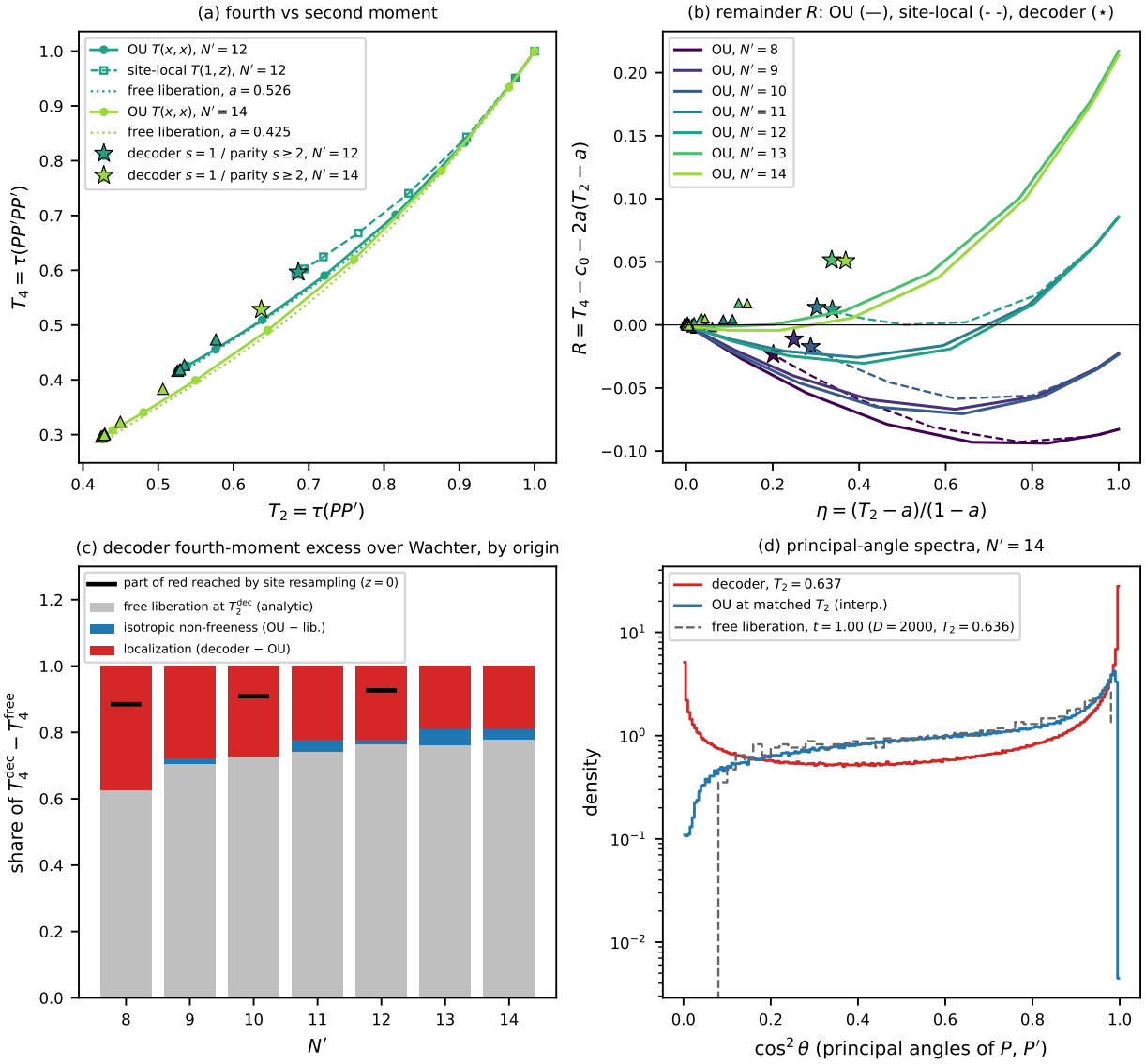


Figure 3: [numerical] Decoder vs isotropic and site-local deformations.

- (a) T_4 against T_2: OU family (solid), site-local family (dashed), free liberation (dotted), decoder ($\star$) and parity probes (▲), at $N' = 12, 14$.
- (b) $R = T_4 - c_0 - 2a(T_2 - a)$ against $\eta = (T_2 - a)/(1 - a)$ for all N'. The sign of the OU curves flips between $a > \frac{1}{2}$ and $a < \frac{1}{2}$, as R_{lib} predicts.
- (c) The three-way split of Table 4. Black ticks: how much of the localization (red) the site resampling $z = 0$ reaches.
- (d) Principal-angle spectra at $N' = 14$: the decoder, OU at matched T_2 (histograms interpolated between neighbouring lags), and finite-D free liberation ($D = 2000$) at $t = -\ln \eta$. OU and liberation coincide in the bulk; the decoder is dominated by wall pile-ups at $\cos^2 \theta \in \{0, 1\}$.

	excess of T_4 over the OU curve at equal T_2				small-angle κ			
N'	$z = 0.707$	$z = 0$ (resample)	$z = -1$ (decoder)	$z = 0$ share	$1/(1-a)$	OU	site	site/free
8	0.0021	0.0121	0.0174	0.69	4.38	4.51	5.36	+22 %
10	0.0039	0.0213	0.0324	0.66	2.80	3.02	3.86	+38 %
12	0.0057	0.0279	0.0414	0.67	2.11	2.36	3.15	+49 %

Table 5: [numerical] The site-local family $T(1, z)$. Its $z = -1$ endpoint reproduces the decoder's excess (Table 4) within errors. Gaussian resampling of only the site's couplings ($z = 0$, no $\mathbb{Z}_2$ sign) already gives two thirds of it. The excess grows smoothly from the identity, $\propto (1 - T_2)^2$ near $z = 1$, as (9) requires. κ is evaluated at the point nearest the identity: $x = 0.981$ (OU) and $z = 0.924$ (site).

8 What R is, and what a gravity computation should target

8.1 Decomposition of R

At equal T_2,

$$R_{\rm dec} = \underbrace{R_{\rm lib}(\eta)}_{\text{analytic}} + \underbrace{\delta_{\rm iso}}_{R^{\rm OU}(x_*) - R_{\rm lib}} + \underbrace{\delta_{\rm loc}}_{T_4^{\rm dec} - T_4^{\rm OU}(x_*)} . \tag{13}$$

At $N' = 12$ ($\eta = 0.338$) the three terms are -0.024, -0.005 and $+0.042$, summing to $R_{\rm dec} = +0.012$. The 29 September R was opaque because it is a small difference of a large negative free-probability piece and a large positive local piece. The two closures tried then (size-diagonal characters; a coherent mixture) were closures of this sum; neither separates the pieces.

8.2 Mechanism of the localization term [interpretation], grounded in §5

- The decoder weight of a chaos component is $1 - 2o(\alpha)/N$; isotropic OU weighs it $x^{|\alpha|}$. The two agree when cross-chord index sets are independent.
- In $\mathcal{N} = 2$ SYK, $\{\psi_J, \psi_K^\dagger\} = 0$ for disjoint J, K, so the chaos content of P is built from overlapping index-set pairs ($P_{[1]} \propto H^{(2)}$). Chords through site i therefore cluster. The number of probe-sensitive chords is then overdispersed relative to an isotropic deformation. At fixed mean (equal T_2) this raises $E \sin^4\theta$.
- Spectrally this is the wall pile-up: BPS states in which site i is effectively frozen or maximally uncertain. It is the UV imprint of n_i on the LMRS operator Pn_iP, cf. Chen et al.'s Fig. 5a [12]. Infinitesimally it is the excess heterogeneity κ of the site-restricted quantum metric.

8.3 Gravitational reading [interpretation]

- **Universal part.** The isotropic four-arc function is the overlap of mismatched two-sided BPS wormholes. At $N' \le 14$ ($q \approx 0.1$–0.3) it is close to free liberation.
  - [conjecture] The double-scaled isotropic function is a q-deformed liberation that reduces to the free one as $q \to 0$. Its departure from free is set by the two-model crossing weight $q_{12} = q$.
  - A dynamical reading: *generic coupling changes rotate the BPS space like a free unitary Brownian motion.* This is a dynamical form of the Haar-orientation (BPS-chaos) hypothesis of Belin–Fu–La Rocca [14].

- **Local part.** It vanishes in strict double scaling (a single-site probe trivializes; for heavy probes decoder $\equiv$ OU). It is a finite-p, site-resolved effect that neither structureless matter chords (BINT/BLY) nor super-JT capture.
  - The exact zero-energy $\mathcal{N} = 2$ OTOC now available (G. Lin, eqs. 5.26–5.38 [11]) is the triple-scaled limit of the isotropic, universal part at best.
  - LMRS's large-Δ suppression of crossed diagrams (their eq. 159 [13]) is irrelevant for our light probe.
- **Consequence for 7.3.2.** The chord computation should target $T_4^{\mathrm{OU}}(x)$, which is universal and liberation-like, with the localization treated as a separate finite-p correction. A single "$\mathbb{Z}_2$ flag per chord" calculation is not expected to reproduce the decoder's R at $p = 3$.

exact finite-N object	chord / gravity counterpart	status
chaos degree n of $P(\omega)$	BLY chord number (n XO pairs)	exact; $\pi_0 = a$; $\pi_1 \to$ BLY
$\operatorname{supp} J \subset \{k \le (p-1)n\}$	BLY size$_O = p\, n_O$	exact; shift $p \to p-1$ (friend mode)
$\kappa_n = a\, p_n(0)^2$	$P_n = a_\lambda p_n^{\mathrm{DS}}(0)^2$	exact both; $\kappa \approx \pi$
$\hat{\chi}_2(U)$ on chord 1	$q^{2\Delta}$ per chord pair	exact vs double-scaled
OU correlation x	matter weight $\tilde{q}$ ($x^{\hat{N}}$)	exact (BKS, Pluma–Speicher)
$T_4^{\mathrm{OU}}(x)$	overlap of mismatched BPS wormholes; q-OU OTOC	structural (double scaling)
T_4^{OU} vs T_2^{OU}	free liberation ($q \to 0$?)	numerical, deviation ≤ 0.012
decoder $-$ OU at equal T_2	site-resolved chord clustering	numerical; absent in DS
$R_{\mathrm{lib}}(\eta)$	free Jacobi process	exact, closed form

Table 6: Dictionary update. It replaces the rows "size spectrum $\leftrightarrow P_n$: empirical" and "decoder R: open" of the 29 September Table 6.

8.4 Next computations

1. **Analytic model of the localization term.** Start from (9) restricted to the couplings through site i, i.e. the variance of the site-restricted quantum-metric operator. Then extend to finite angle along the site family.
2. **Two-colour chord computation of** $T_4^{\mathrm{OU}}(x)$ in BBNR/BLY double scaling (the mismatched-wormhole overlap). Test "q-deformed liberation": $q \to 0$ should give the free Jacobi curve.
3. $N' = 15$–**16** with the BLAS3 basis, for the large-N' trend of the localization share.
4. **Finite-N HH recursion for π_n,** $n \ge 2$ (open since 2 October): the chaos projections of $QP = PQ = 0$.

9 Open questions and caveats

1. **OU $\approx$ liberation is empirical.** It holds at $N' \le 14$, i.e. at small q, with deviations growing with q. The OU curve cannot equal liberation exactly (non-analytic $\eta^2 \ln \eta$). "q-deformed liberation" is a conjecture.

2. **No analytic model of the localization term yet.** Its mechanism (chord clustering) is an interpretation supported by the exact weight formula, not a calculation. Its large-N' fate is open. The share falls, the absolute size appears to saturate, and in strict double scaling it should vanish.

3. **Two-sided reading.** It is derived for the double-scaled Majorana q-Gaussian algebra. The $\mathcal{N} = 2$, BPS-projector version is by analogy.

4. **Chaos coefficients of T_4.** The θ_n are not reliably determined by 9 lags (alternating, large). Only the curves are used.

5. $N' = 14$ **statistics.** 14 OU samples, using 4 of the 16 pairs per lag. The site family stops at $N' = 12$.

6. **Liberation is a $D \to \infty$ model.** Finite-D (Weingarten) corrections are $O(D^{-2})$ and negligible here, but the free Jacobi density for general a is only simulated, not computed in closed form.

7. **Superseded.**

   - "$w_{2n} \leftrightarrow P_n$ is an empirical match": replaced by §4.
   - "Per-chord weights factorize in double scaling": replaced by the character $\hat{\chi}_2$.
   - "Correlated couplings need a correlated mixed q-Gaussian beyond Bożejko–Speicher": wrong, see §3.
   - "Two-copy irrep content of R" (29 September item 7.3.3): deprioritized, since R is now understood as a sum of two pieces with different origins.

A Derivation details

A.1 Mehler formula for a diagonal correlation

For a single complex Gaussian coordinate with $\tilde{c} = kc + \sqrt{1-k^2}c'$, the Hermite polynomials $H_{m,n}(c, \bar{c})$ satisfy $E[H_{m,n}(\tilde{c}) \,|\, c] = k^{m+n} H_{m,n}(c)$. For $|k| = 1$ this is the parity $H_{m,n}(-c) = (-1)^{m+n} H_{m,n}(c)$. Products over coordinates give $E[F(\tilde{C}) \,|\, C] = \sum_\alpha K^\alpha F_\alpha(C)$. Pairing with $F(C)$ through the bilinear form $\mathcal{B}$ and using orthogonality of distinct chaoses gives (7).

A.2 Small-angle expansion

Differentiate $P^2 = P$:

- to first order, $P\partial_I P\, P = 0$;
- to second order, $P\partial_I\partial_J P\, P = -P(\partial_I P \partial_J P + \partial_J P \partial_I P)P$.

Hence $P(P_x - P)P = -B^\dagger B + O(\delta\omega^3)$ with $B = (1-P)\left(\sum_I \delta\omega_I \partial_I P\right) P$. The mean shift of $\delta\omega$ is $O(\epsilon)$ and odd fluctuation moments vanish, so corrections are $O(\epsilon^3)$. With $E\,\delta\omega_I \delta\omega_J = 2\epsilon\,\delta_{IJ}$, Wick's theorem on $\mathrm{Tr}(B^\dagger B)^2$ gives the three orderings $IIKK$, $IJIJ$, $IJJI$. Writing $G_{IJ} = g_{IJ} + F_{IJ}/2$, the last two sum to $2\sum \mathrm{Tr}\, g_{IJ}^2$: the $\mathrm{Tr}\, F^2$ terms cancel and the cross terms vanish by symmetry.

A.3 Liberation closed form

With $\lambda = 1$ and $\theta = a$, DHH eq. (2) reads

- $n = 1$: $\dot{m}_1 = -m_1 + a$;
- $n = 2$: $\dot{m}_2 = -2m_2 + 2am_1(2 - m_1)$.

Insert $m_1 = a + bu$. The u^2 term of the source is resonant with the homogeneous solution, which produces the $t\,u^2 = -u^2 \ln u$ term. Fixing $m_2(0) = 1$ gives E.

B Literature added in this cycle

Notes are in `literature/notes/`, and `literature/synthesis.md` §8 compares the tools. Each entry gives what the paper establishes and how it is used here.

- **BBNR [4].**
  - $\mathcal{N} = 2$ double-scaled SYK: oriented chords, friend/enemy weights, the fixed-charge density, and the finite-N ground-state count (our d).
  - The double-chord rule $\tilde{q}^{\hat{N}}$.
  - Four-point functions are listed there as open.
- **BINT [5].**
  - The matter-chord rule, the 2-point vertex (= q-Mehler kernel), and the crossed 4-point function with the $U_q(\mathfrak{su}(1,1))$ R-matrix.
  - Its random-matter rule differs from ours (§5).
- **Pluma–Speicher [6].**
  - Independent SYK copies → q-Gaussians; Brownian couplings → q-Brownian motion by multilinearity.
  - Fluctuations at scale $\binom{N}{p}^{-1/2}$.
- **Bożejko–Kümmerer–Speicher [7].** Second quantization $\Gamma_q(T)$; the q-OU process is Markov; classical versions reproduce only time-ordered correlators.
- **G. Lin [11].** The exact zero-energy $\mathcal{N} = 2$ super-JT OTOC for equal Δ, and the Berry-curvature second moment as TOC−OTOC.

C Reproducibility

All heavy runs were under `scripts/memwatch.py`; the peak was 1.43 GB (at $N' = 14$).

Item	Code / data
Chord/size dictionary (§§3–4, Fig. 1)	`src/chord_chaos.py`, `src/plethysm.py`, `scripts/run_chord_chaos.py`, `analyze_chord_chaos.py`, `run_topsize_projection.py`, `run_chord1_sizes.py`, `plot_chord_size_dictionary.py`; data `results/data/chord_chaos_2026-10-01/`; tests `tests/test_chord_chaos.py`
OU / site / decoder (§7, Fig. 3)	`scripts/run_ou_fourarc.py`, `analyze_ou_fourarc.py`, `plot_ou_fourarc.py`; data `results/data/ou_fourarc_2026-10-05/` (incl. `analysis.json`, `analysis_output.txt`); tests `tests/test_ou_fourarc.py`
Notes	`research/notes/chord_size_dictionary_2026-10-02.md`, `research/notes/decoder_T4_fourarc_2026-10-05.md`; `docs/derivations.md` D4, D5

Seeds and samples. $p = 3$ and $F = \lfloor N'/2 \rfloor$ throughout.

- **Dictionary campaign:** seeds $1000003(s_0 + s) + 17$.
  - 4000/744/174 samples at $N' = 8/10/12$ ($K = 16$);
  - 5 samples at $N' = 14$ ($K = 12$);
  - top-size projections with 40 realizations (seed 41).
- **Four-arc campaign:** seeds $1000003(s_0 + s) + 29$; $s_0 = 0$ for OU and 1000 for the site family.
  - Probe sets use seed+7.
  - $N' = 14$ used the BLAS3 (Gram-eigh) basis and 4 pair starts per lag. That basis is identical to the Hodge/SVD basis to 3×10^{-15}.
- **Code states:** data generated at `ee619d3` (dictionary) and `1aa3dd3` plus working tree (four-arc), recorded in `meta_*.json`. This report is at `4395248` plus this file.

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